PlayCast PRO

Church Presentation & Broadcast Studio Engine

Document ID: ARCH-PLAYCAST-2026-v2.0
Target: Client Presentation & Technical Proposal

✓ PRODUCTION VERIFIED

OFFLINE SCRIPTURE CANON	DUAL-SYNC LATENCY	RECURRING LICENSE FEES
31,102 Verses	**< 2ms / 300ms**	**$0 / Free Forever**
Complete 66 Books in Tamil & English	BroadcastChannel + SQLite Hub	100% On-Premise Church Hardware

1. High-Level System Architecture & Hardware Topology

PlayCast Studio unifies live presentation and streaming into a single, cohesive on-premise system. The desktop operator station controls 4 concurrent outputs simultaneously:



Figure 1.0: PlayCast Multi-Display System Architecture & Physical Hardware Topology

Output Target	URL Route	Visual Format	Primary Church Function
Operator Console	`/`	Dark Broadcast UI	Controls worship song slides, Tamil Bible searches, preview cues, and live cuts.
OBS Live Stream	`/display`	100% Transparent Alpha	Renders lower-third Tamil/English lyrics and Scripture directly over camera video.
Sanctuary Projector	`/projector`	16:9 Fullscreen Canvas	Projects high-resolution 1080p presentation text for the in-person congregation.
Pastor Stage Monitor	`/stage`	High-Contrast Yellow/White	Confidence TV displaying current verse, next slide preview, and live countdown timer.
Mobile Remote	`/remote`	Touch Handheld Web UI	Enables wireless slide advancement from any pulpit tablet or phone over Wi-Fi.

2. Five-Step Operational Presentation Workflow

The core workflow enables church media teams to search, stage, and broadcast any Bible verse or song verse within seconds during live worship:

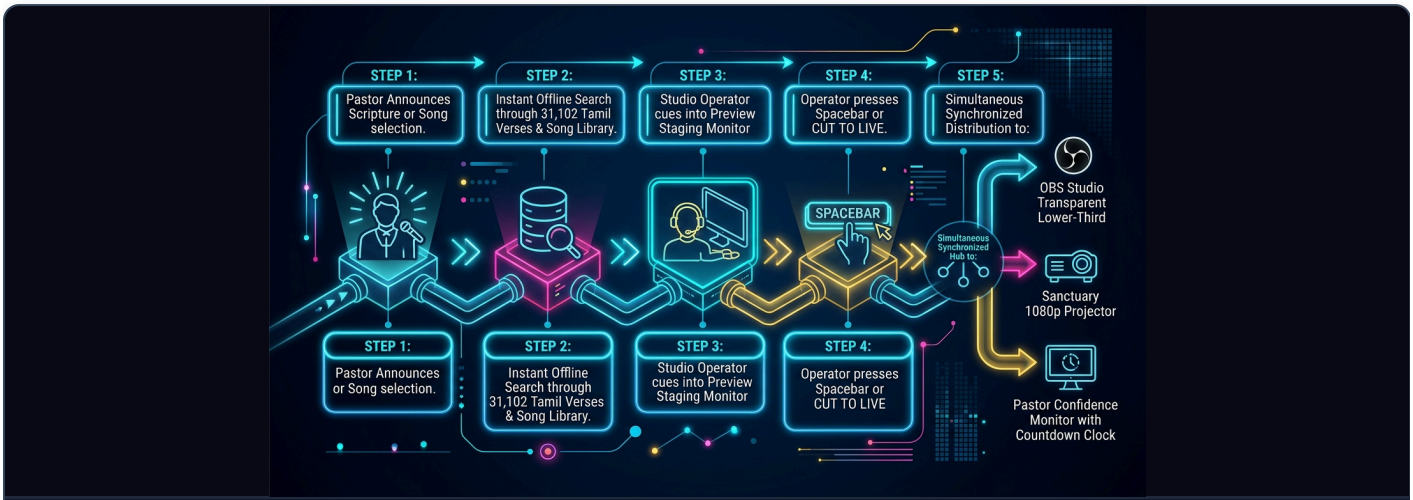


Figure 2.0: Five-Step Live Presentation, Scripture Pulling & Synchronized Distribution Pipeline

1. **Pastor Announces Scripture or Song:** The speaker calls out a reference (e.g., John 3:16) or worship leader starts a song.
2. **Instant Offline Search:** The engine queries 31,102 Tamil verses and song databases in < 10ms with zero internet dependency.
3. **Preview Staging:** The operator selects the slide, cueing it into the Studio Preview Monitor with cyan highlight.
4. **Live Cut Transition:** The operator presses Spacebar or clicks "CUT TO LIVE", triggering cross-dissolve animations.
5. **Synchronized Multi-Screen Output:** The slide broadcasts simultaneously to OBS Studio (transparent lower third), the Sanctuary Projector, and the Stage Confidence Monitor.

3. Commercial Advantage & Competitive Client Comparison

When pitching to church leadership, PlayCast provides key technical and financial advantages over costly legacy commercial systems:

Evaluation Metric	Commercial Tools (ProPresenter / EasyWorship)	PlayCast Studio Solution
Licensing & Costs	$399 – $999 upfront + $180/year renewal per computer.	**$0 recurring fees.** 100% self-hosted on church hardware.
Tamil Unicode Engine	Prone to broken Tamil glyphs and rendering misalignment.	**Native Tamil font rendering.** Full 31,102 verses pre-indexed.
OBS Studio Streaming	Requires expensive NDI plugins or virtual capture cards.	**Direct Browser Source URL.** Plug-and-play transparent overlay.
Hardware Footprint	Requires heavy gaming GPUs and dedicated workstations.	**Ultra-lightweight.** Runs on standard laptops via portable PHP.
Stage Confidence Display	Costly add-on licenses or separate hardware boxes.	**Built-in Stage TV Web App** with Next-Slide preview & timers.
Internet Dependency	Cloud license checks fail if internet drops during service.	**100% Offline-First.** Functions completely disconnected from the web.

> **Client Proposal Summary:** PlayCast delivers high-end broadcast capabilities, professional multi-window outputs, and native Tamil Bible integration without high commercial software licenses or recurring subscription lock-in.